

Retail Demand Forecasting

Process Flow Document

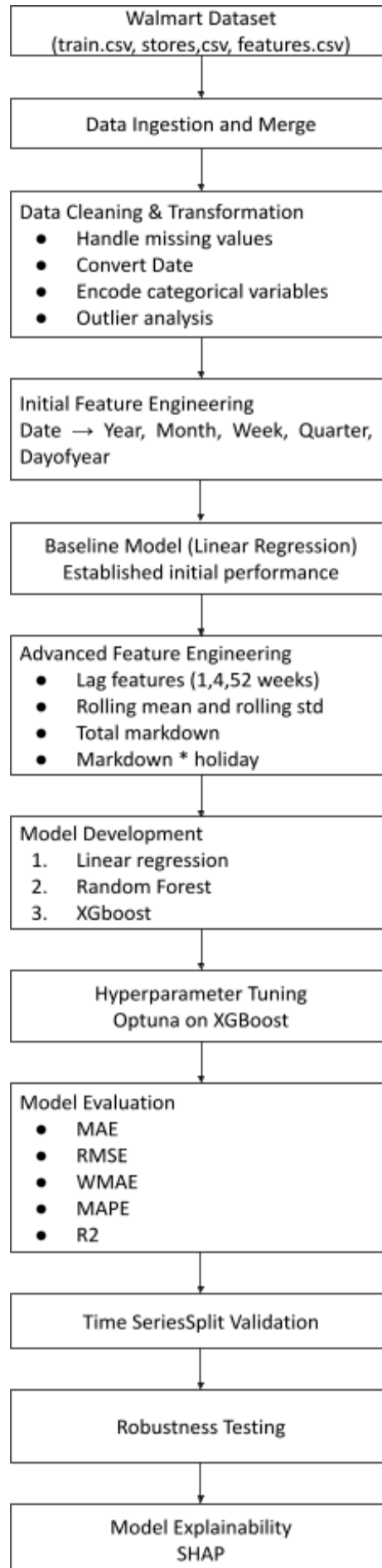
Technical Assignment

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1. Data Ingestion

- Loaded train.csv, features.csv and stores.csv
- Converted Date column to datetime.
- Merged train with features
- Merged resulting dataset with stores.
- Used left join so no sales rows get dropped, just added extra info to them.

2. Data cleaning & transformation

- Checked missing values
- Missing values in Markdown1-Markdown5 were imputed with 0, assuming no promotion was recorded for those weeks.
- Created missing value indicator (flag) features for the markdown columns to preserve information about originally missing entries.
- Mapped holiday dates to create the Holiday_Name feature.
- Flagged outliers using the IQR method on log-transformed sales for analysis.

3. Basic Feature Engineering

- Created calendar features like Year, Month, Week, Quarter, DayOfYear
- Purpose is to capture seasonality and temporal sales patterns.

4. Baseline Model

A baseline Linear Regression model was trained using the initial engineered features to establish a benchmark. Performance was evaluated using MAE, RMSE, WMAE, MAPE and R2.

The baseline provided a reference for measuring improvements after advanced feature engineering and hyperparameter tuning.

5. Advanced Feature Engineering

To improve the model's ability to capture sales trends and promotional effects, additional features were created.

Lag Features

- Sales_lag_1 - Sales from previous week.
- Sales_lag_4 - Sales from four weeks earlier
- Sales_lag_52 - Sales from the same week in the previous year.

Rolling Features

- Sales_roll_mean_4 - Average sales over the previous four weeks.
- Sales_roll_std_4 - Standard deviation of sales over the previous four weeks.

Promotion Features

Total_markdown - Sum of all markdown values.

Markdown x Holiday - Interaction between promotional markdowns and holiday weeks.

Purpose - Model understands the combined impact of promotions and holidays on sales.

6. Model Training and Validation

Data Splitting

Used an 85:15 chronological split to separate training and validation data.

I used a time-based split instead of random, since randomly mixing future and past rows would let the model cheat by seeing future data early.

Models trained

Linear regression

Random forest

XGBoost

Model Evaluation

The models were evaluated using the following metrics:

MAE - Measures average prediction error.

RMSE - Penalizes larger prediction errors.

WMAE - Gives 5x higher weight to holiday weeks.

MAPE - Measures percentage error.

R2 Score - Indicates how well the model explains sales variability.

Hyperparameter Tuning

Applied Optuna to optimize the XGBoost model.

Tuned parameters included learning rate, maximum depth, number of estimators, subsample ratio, column sampling, minimum child weight, and regularization parameters.

	Model	Train MAE	Validation MAE	Train RMSE	Validation RMSE	Train WMAE	Validation WMAE	Train MAPE	Validation MAPE	Train R2	Validation R2
0	Linear Regression	1672.387	1558.881	3709.816	2943.168	1800.551	1606.646	28.492	37.538	0.974	0.982
1	Random Forest	1046.089	1214.999	2279.664	2552.978	1085.186	1251.307	12.611	14.069	0.990	0.986
2	XGBoost	954.777	1239.832	1646.155	2655.579	964.315	1275.661	14.256	15.599	0.995	0.985
3	XGBoost (Optuna)	1045.297	1209.714	1940.723	2536.662	1053.777	1250.612	14.352	14.919	0.993	0.987

The tuned XGBoost model achieved the best validation performance and was selected as the final model.

7. Time Series Validation

Performed TimeSeriesSplit on the final tuned XGBoost model.

This ensured the model was always trained on past data and validated on future data, confirming its robustness for forecasting.

	model	MAE	RMSE	WMAE	MAPE	R2
0	Fold 1	1819.423	4607.376	1951.749	16.876	0.970
1	Fold 2	1496.048	3278.608	1595.135	16.664	0.977
2	Fold 3	1533.122	3247.504	1612.902	15.277	0.978
3	Fold 4	1097.831	2743.006	1188.630	23.204	0.978
4	Fold 5	1121.653	2529.831	1166.074	15.838	0.984

8. Robustness Testing

Evaluated the tuned XGBoost model's performance across holiday weeks, store types and markdown levels to check for consistency.

Holiday vs Non-holiday weeks

- Holiday vs Non-Holiday weeks
- The model performs slightly better during non-holiday weeks.
- Because holiday sales are more volatile and difficult to predict

Store Type Analysis

- Evaluated performance across Store Types A,B,C
- The model generalized well across all store types.
- Store type C achieved the lowest prediction error, while Store Type A showed slightly higher errors due to large sales variability.

Markdown Level Analysis

- Sales records are grouped into low, medium and high markdown categories.
- The model remained accurate across all promotional levels.
- Prediction errors increased slightly for high markdown periods, indicating that large promotions introduce additional uncertainty in customer demand.
- Overall the model help up well across holidays, store types, and promotion levels.

9. Model Explainability

Used SHAP (TreeExplainer) to interpret the tuned XGBoost model's prediction for business stakeholders.

Global Interpretability

Generated SHAP summary bar chart on a 3000 row sample of the validation set to identify the most influential features.

Sales_lag_1 and sales_lag_52 were the two most important features by a wide margin, followed by Sales_roll_mean_4.

External features (CPI, Unemployment, Temperature, Fuelprice) have minimal impact, confirming the model relies primarily on recent sales history and yearly seasonality rather than economic indicators.

Local Interpretability

Generated SHAP waterfall plots for 3 individual validation predictions - highest error, lowest error and a random example, to show how the model arrives at specific forecasts.

Lowest error (Actual: 154.06, Predicted: 154.05): Near-perfect prediction for a low volume stable store-dept combination, where recent sales lag features closely matched actual demand.

Random example (Actual: 6390, predicted: 6182): Prediction within 3% of actual sales, it represents typical model performance under normal conditions.

Highest error (Actual: 114230, Predicted: 45542): Occurred during a holiday week with a large active markdown (27000), so the model underpredicted because the previous year's same week had much lower sales from sales_lag_52, causing it to underestimate the combined lift from the holiday and promotion.

Business Interpretation

Which factors matter most

Recent sales history (last week's sales) and yearly seasonality (same week last year) are by far the strongest drivers of weekly sales, far more influential than external economic conditions like CPI, unemployment or fuel prices.

So sales mostly follow recent trends and last year's pattern, the economy barely matters.

So during big sales or promotions expect the forecast to be less reliable.

How stores should prepare for holiday

Since holiday weeks show higher prediction error and higher sales volatility, store teams should treat model forecasts for holiday weeks as a starting estimate rather than a precise number, and build in buffer inventory/staffing for major holidays (Thanksgiving, Christmas, Labor Day, Super Bowl) rather than relying solely on the model's point forecast.

Role of markdowns: Markdowns clearly boost sales, but they also make sales harder to predict accurately, prediction error increases as markdown intensity increases. This suggests that during large promotional events, business teams should expect more forecast uncertainty and may want to monitor actual sales closely during the promotion rather than fully trusting the pre-promotion forecast.

Combined effect: The model's biggest blind spot is when holidays and markdowns overlap, this is when demand spikes are largest and hardest to predict, since the model leans on historical patterns that don't fully anticipate the combined lift. This is the scenario where manual review or added safety stock is most valuable.

Final Insights

Started with a basic Linear Regression model that barely explained anything (R^2 score of 0.09), and ended with a tuned XGBoost model that explains 98.7% of sales variation.

Feature engineering (lag and rolling sales features) made the biggest difference, more than the model choice itself.

The model mostly relies on recent sales and last year's sales for the same week to make predictions. Things like weather, fuel prices, or unemployment barely matter - sales are driven much more by short term trends and seasonality than the economy.

The model stays consistent over time. When tested across different time periods, it kept performing well and even improved as it had more history to learn from.

The model is slightly less accurate during holiday weeks and high-markdown periods, these are naturally harder to predict since sales can spike suddenly. One example: during a holiday week with a big promotion, the model badly underestimated sales because it leaned too much on last year's numbers, which happened to be much lower.

Overall, the model is solid for everyday forecasting, but for holiday or big-promotion weeks, it's a good idea to double-check the numbers manually rather than relying on it fully.